

Mohammad Joghataee

Machine Learning Scientist · Medical NLP, Clinical Notes, and Foundation Models

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PROFESSIONAL SUMMARY

Ph.D. candidate in computer science building machine learning and generative AI systems for clinical text and EHR data. Fine-tunes open-source foundation models to summarize and extract themes from unstructured notes, designs HIPAA-safe NLP pipelines, and publishes research-grade clinical models that clinicians can inspect. Comfortable moving from messy hospital data to a prototype people will actually use, then writing down what the model cannot be trusted to do.

CORE SKILLS

Medical NLP	Clinical note summarization, theme / evidence extraction, PII removal, biomedical relation extraction, knowledge graphs
ML / Foundation models	PyTorch, TensorFlow, scikit-learn, open-source LLMs, prompt design, fine-tuning, SHAP, calibration, ablation
Clinical data	Multi-site hospital EHR, observational analysis, missingness, HIPAA / IRB, clinician review of model output
Build	Python, SQL, reproducible pipelines, research-to-prototype delivery, Git, Linux

RESEARCH EXPERIENCE

Research Assistant — Clinical ML & Generative AI | Auburn University · Mayo Clinic collaboration 2022 – Present

Research on medical NLP and EHR models that have to survive clinician scrutiny, not only a test split.

- Built a HIPAA-compliant generative AI pipeline using prompt design and fine-tuned open-source LLMs to summarize unstructured clinical notes from rare-disease patients and extract themes for clinician review.
- Wrote an automated NLP tool to detect and strip personally identifiable information from clinical notes so note models could run under IRB and HIPAA constraints.
- Trained and validated EHR models on 11,000+ multi-site hospital encounters (180+ features) for rare and atypical infections; reported calibration (Brier 0.057), sensitivity limits, and site-shift risk instead of claiming a screening product.
- Used SHAP and ablation studies so clinical collaborators could see which evidence supported a score; iterated after clinician feedback rather than freezing a leaderboard metric.
- Shipped research outputs into clinician-facing dashboards and peer-reviewed papers (International Journal of Medical Informatics; Biomol Biomed; JAMIA under review).
- Worked with Mayo Clinic teams on ICU time-series / activity models, aligning labels, reconciling messy sensor data, and keeping methods reproducible.

NLP Researcher — Biomedical Text | Iran University of Science and Technology 2019 – 2021

- Researched biomedical relation extraction and knowledge-graph construction from scientific text; built cleaning, entity linking, and model-deployment pipelines.

Teaching Assistant — Software Engineering & Databases | Auburn University 2022 – Present

- Taught software engineering and database systems; mentored students on Python, SQL, and turning a prototype into something another person can run.

EDUCATION

Ph.D. Candidate, Computer Science and Software Engineering Aug 2022 – Present

Auburn University · GPA 3.93/4.0 · Clinical NLP and EHR models for rare-infection detection · Advisors: Dr. Ashish Gupta, Dr. Xiao Qin

M.S., Management of Information Systems 2023 – 2025

Harbert College of Business, Auburn University · GPA 4.0/4.0 · Certificates: Business Analytics, Cybersecurity

M.S., Computer Science and Engineering 2019 – 2021

Iran University of Science and Technology · GPA 18.16/20 · Biomedical NLP

PUBLICATIONS

- Channel-Recalibrated Temporal Deep Learning for Interpretable Classification of PTSD and Post-Concussive Syndrome Using Resting-State fMRI. *Journal of the American Medical Informatics Association (JAMIA)* — Under Review, 2026.
- Machine Learning Approaches to Identify Rare and Atypical Infections in Hospitalized Patients. *International Journal of Medical Informatics*, 2026.
- Development and Validation of a Preliminary Multivariable Diagnostic Model for Identifying Unusual Infections in Hospitalized Patients. *Biomol Biomed*, 2024.

- History of Multiple Concussions Associated with Impaired Perceptual Decision-Making Metrics Identified by Both Machine Learning and Statistical Regression Methods. *PLOS One*, 2026.
- Multimodal Approaches for Classifying PTSD and PCS-PTSD from Resting-State fMRI: Deep Learning on Time-Series and Interpretable Feature-Based Models. *Pre-ICIS SIGDSA Symposium Proceedings*, 2025.
- Predicting PTSD Severity in Veterans: An Explainable fMRI-Based Deep Learning Approach. *Pre-ICIS SIGDSA Symposium Proceedings*, 2025.
- Development of a Body Movement Classification Algorithm Using Wearable Accelerometers: A Pilot Study. *AMIA*, 2024.
- Modeling Long COVID Disease Network in Pediatric Population. *AMCIS TREOs*, 2023.
- AI Models for Rare Infectious Disease Detection. *Decision Sciences Institute*, 2023.